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GeoPops: A Python package for generating geographically-realistic synthetic populations and networks

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Software

- Review
- Repository
- Archive

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Summary

GeoPops (Hamilton, Kerr, et al., 2025) is an open-source Python package that generates synthetic populations for any US Census area from publicly available data. GeoPops 1) creates a set of agents approximating US Census population distributions, 2) assigns agents to homes, schools, workplaces, and group quarters facilities (e.g., nursing homes) with spatial dependencies, and 3) connects agents within these locations using network algorithms. The generated output files include a list of agents with attributes (age, gender, income, race/ethnicity, and more) and network structures detailing connectivity between agents. GeoPops is designed for infectious disease modeling but can be applied to a variety of public health issues (e.g., disaster response, education policies) and combined with other types of connectivity networks (e.g., roads, sewage). A GeoPops population is ideal for scenario modeling using agent-based models and networks, and can also be used with compartmental models and contact matrices. Users can make their own county, state, or multi-state populations. The GeoPopsHub on GitHub serves as a central location for source code, tutorials, and community contributions.

Statement of need

Synthetic populations that spatially connect individuals and capture activity-based interactions (e.g., going to school) using networks are useful for modeling infectious diseases, particularly when assessing the impact of interventions on geographic and demographic subgroups. Several synthetic population packages exist that generate agents within households for US Census locations but do not generate activity networks. GeoPops is presently the only one with geographically-coded school and workplace assignment and network generation that is completely open-source and actively maintained. Synthetic populations without pre-built networks require modelers to construct their own, which can be data and time intensive, and limits standardization, consistency, and comparability across models. To support both high-resolution analyses and time-sensitive response modeling, it is essential to improve the accessibility and usability of networked synthetic population tools. This includes providing open-source code, clear documentation, and user-friendly interfaces and tutorials that accommodate a range of programming experience.

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State of the field

We identified ten synthetic population generators for US Census locations (Table 1); however, few are actively maintained with open-source code for activity assignment and network generation. They differ in geographic and demographic granularity and method of agent/household synthesis (e.g., Combinatorial Optimization (CO), Iterative Proportional Fitting (IPF), or hierarchical reconstruction (HR)). Publications for each tool show relative accuracy for each method when comparing synthesized populations to observed Census data. HR is faster when compared to CO and IPF but less accurate at higher granularity. We chose CO as GeoPops' default method because it scales well in terms of accuracy and computational requirements with increasing numbers of target variables (i.e., individual and household characteristics). Future iterations will allow users to select a method depending on their modeling needs and constraints (e.g., speed vs granularity, data availability).

Table 1: Capability comparison of different synthetic population generators: GeoPops (JHU) (Hamilton, Kerr, et al., 2025); Geo-Synthetic-Pop (U. at Buffalo) (Jiang et al., 2024); UrbanPop (Oak Ridge) (Tuccillo et al., 2023); SHAPE (George Mason) (Von Hoene et al., 2026); SynthPop (RTI) (Rineer et al., 2025); Synthetic populations for regions of the World (UVA); SynthPops (Gates); ChiSIM (Argonne) (Macal et al., 2018); synthetic-populations (U. of Chicago) (Lin & Xiao, 2023). *Others available upon request.

	GeoPops	Geo-Synthetic-Pop	UrbanPop	SHAPE	SynthPop	UVA	SynthPops	ChiSIM	synthetic-populations
Baseline population									
Generates agents within households	✓	✓	✓	Agents only	✓	✓	✓	✓	✓
Open-source code	✓	✓	✓	✓	✓		✓		✓
Uses only publicly available data	✓	✓	✓	✓	✓				✓
PyPI/CRAN repository	✓		✓	✓			✓		
Actively maintained	✓		✓	✓	✓	✓			✓
Downloadable pre-built state populations	All states (coming soon)	All states	Some states*	All states	All states	Some states*	Some states		All states
Geographic granularity									
State	✓	✓	✓	✓	✓	✓	✓	✓	✓
County	✓	✓	✓	✓	✓	✓	✓	✓	✓
Tract	✓	✓	✓	✓	✓	✓		✓	✓
CBG	✓		✓		✓	✓		✓	✓
Population synthesis years available									
2019 and earlier	✓		2016–2019	✓	✓	✓	✓	✓	✓
2020 and later	✓	2020 only	Excluding 2020–2022	✓	Upon request	Upon request	?	✓	
Activity networks									
Generates activity networks	✓	✓	Assignment only		✓	✓	✓	✓	
Open-source code	✓	✓					✓		
Uses only publicly available data	✓	✓	✓		?		✓		
Actively maintained	✓		✓		✓	✓		✓	
Downloadable network files	All states (coming soon)	All states	Assignment for some states		All states	Some states*	Some states*		

Software design

GeoPops builds on its predecessor package, GREASYPOP-CO (Tulchinsky et al., 2026), and uses the same methodology but includes several structural changes to increase speed and usability. Data processing code has also been refactored to work with Census data schema

from 2020 and beyond, while maintaining backward compatibility with prior years (Census boundaries and some variable codes changed in 2020). GeoPopsHub includes a detailed tutorial on how to build a population and run a measles model using the example of Spartanburg County, SC.

Data downloading and processing

The user specifies a geographical region, and GeoPops downloads all the necessary data from publicly available sources (Table 2) through an Application Programming Interface (API). During processing, we combine select Census columns to avoid over granular targets in the CO step. For CO, we use 85 target columns that match with PUMS columns and are relevant to disease transmission and useful in generating activity networks. We use IPF to estimate workers by industry by residence type because Census data on employment by CBG does not distinguish individuals living in households versus group quarters. We also use IPF to create origin-destination and industry-destination commute matrices by CBG from LEHD data.

Table 2: Data sources and role in GeoPops.

Data source	Role in GeoPops
American Community Survey (ACS)	Provides Census block group (CBG) level demographic and socioeconomic distributions (e.g., age, households, income) that GeoPops matches when constructing the synthetic population.
Decennial Census	Used to construct group quarters compositions (institutional vs non-institutional, civilian vs military).
Public Use Microdata Sample (PUMS)	Provides person- and household-level microdata that are reweighted and combined (via combinatorial optimization) to create realistic synthetic households and individuals.
TIGER/Line Shapefiles	Provide the geographic boundaries (county, tracts, CBGs) needed to locate agents spatially and to link Census counts to geography.
LEHD LODES (Longitudinal Employer–Household Dynamics)	Provides origin-destination commute patterns and workplace locations, used to assign workers to jobs and build the workplace contact network.
County Business Patterns (CBP)	Supplies employment counts by industry and geography, used to characterize and weight workplaces in the synthetic population and work network.
Census Tract-to-PUMA Crosswalk	Links Census tracts to PUMAs so that PUMS microdata (available at the PUMA level) can be aligned with tract/block-group targets.
MCDC Geocorr (2018/2022)	Provides crosswalks between tracts/CBGs and higher-level geographies (counties, CBSAs, urban-rural), used to attach regional attributes and to aggregate or filter results.
NCES EDGE School Data	Supplies public school locations and characteristics, used to assign students and staff to schools and to build the school contact network.

Combinatorial Optimization

GeoPops uses CO with simulated annealing to generate agents within households. CO heuristically samples real household records from PUMS data until a set is found such that synthesized distributions of individual and household characteristics reasonably match observed marginals from the ACS. The result is a list of agents with attributes (age, gender, occupation, etc.) who are assigned to households within CBG locations (Figure 1).

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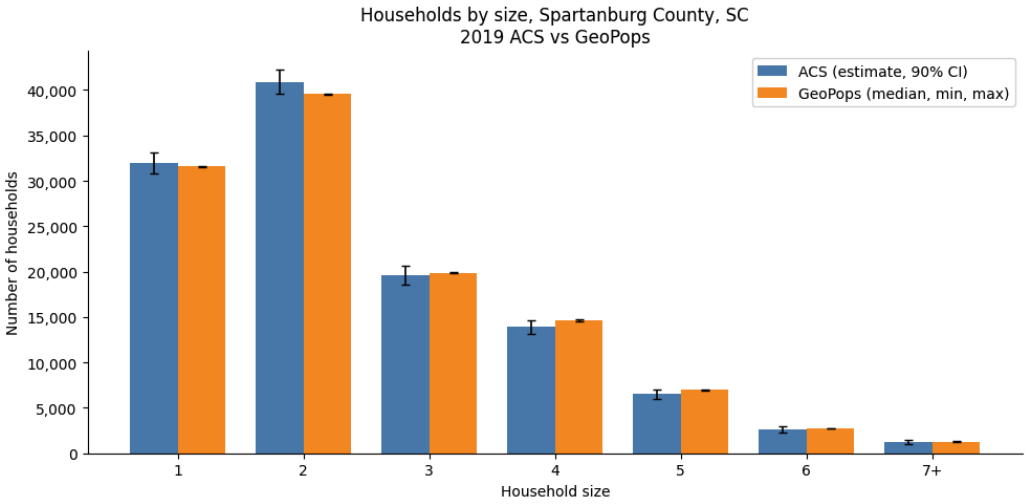


Figure 1: Comparison of ACS and GeoPops for number of households by household size for Spartanburg County, SC. Ninety percent confidence intervals (CI) for ACS data (blue) are calculated using margin of error values provided by the US Census (United States Census Bureau, 2019). GeoPops data (orange) include the median with minimum and maximum range from 100 population synthesis runs. The ranges for household size counts resulting from 100 GeoPops runs fall within the 90% CI ACS estimates, except for 2-person households where the GeoPops maximum overlaps with the ACS lower bound by ~50 households.

Assignment

Students are assigned to the closest school (Euclidean distance from the centroid of the student's home CBG to the school's location from NCES data) with capacity and the required grade levels with 90% probability and to the next closest school with 10% probability. For worker assignment to workplaces, we multiply counts of commuters by industry category in each CBG by industry-destination commute matrices. School workers are assigned to schools by selecting education-industry workers who commute to the same CBG as the school. We consider five categories of group quarters facilities: institutional for ages under 18; institutional, civilian non-institutional, and military for ages 18 to 64; and institutional for ages 65 and over. For a given CBG, the ACS provides estimates of how many people live in each of these categories but does not specify the exact type (e.g., nursing home or incarceration facility). In each synthesized CBG, we create a group quarters facility of a certain category if at least 20 residents are of the appropriate age/category status.

Network generation

GeoPops generates network layers by default; however, users can also create their own networks; for example, we are currently working on a between-school network as a proxy for sports games, etc. Agents are fully connected within each home. Within schools and workplaces, agents are connected using Stochastic Block Modeling (SBM). For schools, the default mean degree is 12 and blocks represent different grades. For workplaces, the default mean degree is 8 and blocks represent income categories. For both school and workplace networks, an assortativity coefficient of 0.9 is used, which means, for example, an average of 90% of a student's contacts will be with other children in the same grade (Figure 2A). Within group quarters facilities, agents are connected using the small-world algorithm with a mean degree of 12 and a rewiring probability of 0.25 (Figure 2B). Users can adjust network algorithm parameters to better reflect their modeling context if needed.

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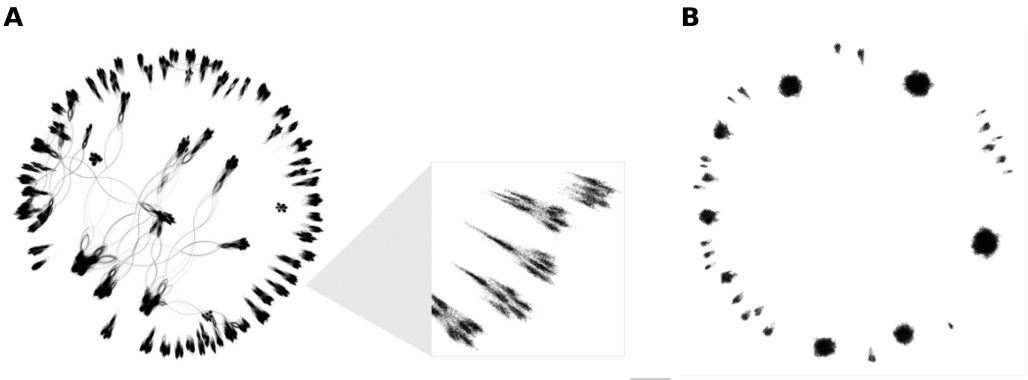


Figure 2: Activity network visualizations for Spartanburg County, SC. A) School network made with SBM. Agents have a mean degree of 12 and are clustered within grades within schools as a proxy for classrooms; the inset shows grade clusters within one school. The plot, created with Gephi (v0.10.1), shows some between-cluster connections. These are caused by agents who are students in one school and workers in another school; for this run, the synthetic population includes 14 agents where this is the case. B) Group quarters network where agents are connected within small-world graphs within each facility.

Running GeoPops

The user creates a parameter dictionary and passes it into the `geopops.run_all()` function, which runs all of the construction steps automatically (see the example below). Alternatively, the user can run each step individually by calling functions one at a time (Table 3). The `geopops.for_starsim` module includes functions to create people and network objects compatible with the open-source agent-based modeling software Starsim (Kerr et al., 2024). Raw data files, interim files, and processed output are all stored locally. This design allows users to jump quickly into agent-based modeling with Starsim, while also enabling more detailed use by experienced programmers. Total runtime for Spartanburg County is ~11 minutes on a 2026 MacBook Pro and household internet, and most of this is downloading data. Runtime for the entire state of South Carolina is ~38 minutes, and for a multi-state population of South Carolina and North Carolina ~99 minutes (Table 4). We will soon provide pre-built state populations for download.

```
import geopops

# Define parameters
pars_geopops = {
    'path': "data",                    # Folder where output files are stored
    'census_api_key': "YOUR_API_KEY", # Your Census API key
    'main_year': 2019,                 # Year of data
    'geos': ["45083"],                 # State or county FIPS (Spartanburg, SC)
    'commute_states': ["45", "37"],    # State FIPS of commute data (SC, NC)
    'use_pums': ["45", "37"],          # State FIPS of PUMS data (SC, NC)
}

# Generate population with .run_all()
geopops.run_all(pars=pars_geopops)
```

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Table 3: Function and module descriptions.

Function/module	Description
`geopops.run_all()`	Takes parameter dictionary as input. Automatically runs all the following steps sequentially (function).
`geopops.write_config()`	Takes parameter dictionary as input. Writes `config.json` with specified parameters (function).
`geopops.download_data()`	Downloads raw data files from Census API and URLs (function).
`geopops.process_data()`	Processes data for population generation (function).
`geopops.generate_pop()`	Builds the population and networks (function).
`geopops.for_starsim`	Exports agent and network files for Starsim (module).

Table 4: Approximate runtimes for county, single-state, and multi-state populations. Single county: Spartanburg County with commute and PUMS data for SC and NC. Single state: SC with commute and PUMS data for SC, NC, and GA. Multi-state: SC and NC with commute and PUMS data for SC, NC, VA, TN, and GA.

GeoPops function	Single county	Single state	Multi-state
`download_data()`	9 min	11 min	18 min
`process_data()`	1 min	1 min	4 min
`generate_pop()`	1 min	26 min	77 min
Total runtime	**11 min**	**38 min**	**99 min**

Research impact statement

GeoPops has been presented at several research conferences, including the 2026 *Insight Net Annual Meeting* (Hamilton et al., 2026), the 2025 *Winter Simulation Conference* (Hamilton, Tulchinsky, Lin, et al., 2025), *Epidemics* in 2025 (Hamilton, Du, et al., 2025), and the 2025 *MIDAS Network Annual Meeting* (Hamilton, Tulchinsky, Kerr, et al., 2025). The GeoPopsHub currently has 15 GitHub members. In addition to development collaborators, researchers at other institutions are now using GeoPops, including the Johns Hopkins University School of Public Health, the University of Utah, and Duke University. GeoPops is also being used to teach simulation modeling for public health decision making in an undergraduate course for Johns Hopkins engineering students.

AI usage disclosure

The software Cursor (v2.7–3.7) and VS Code (v1.135.0) was used in the development of GeoPops starting with the GREASYPOP-CO source code. For example, an AI agent was used to translate R code into Python for data downloading and organize previous scripts into new functions. Often, the agent's first pass was unsuccessful, in which case the chat feature was used to assist the debugging process. AI agents were also helpful in generating comments and doc strings and exploring raw Census data and metadata PDFs to determine where variable codes had been changed from 2019 to 2020. Claude Code (Opus 4.5) was used to port the Julia code for CO into Python. ChatGPT was occasionally used for more general software development questions.

Code and data availability

All source code and tutorials are available on GitHub (Hamilton, Kerr, et al., 2025). Soon we will publish pre-built state populations.

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- One Health Trust (Design)
- Johns Hopkins University Applied Physics Laboratory (Design)

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